

TRAVELTIDE SEGMENTATION ANALYSIS - TECHNICAL DOCUMENTATION

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Purpose: Validate Elena's 5 rewards hypotheses (Flight/Hotel Discounts, Cancellation Flexibility, Baggage, Weekend Travel) by classifying 6,000 active users into behavioral segments.

Methodology:

1. user_filter: Active users with >7 sessions since Jan 4, 2023.
2. user_segments_parameter: Aggregates per-user KPIs (avg nights, seats, discount usage, cancellation counts, baggage).
3. segment_detail: Rule-based CASE statement mapping users to personas.
4. Final SELECT: Aggregates segments into the summary matrix.

Logic Verification (Edge Cases):

- "NEW": Users with zero completed trips OR all trips cancelled.
 - "BUSINESS": Solo traveler, short stay (≤ 3 nights), booked < 15 days out.
 - "FAMILY": Married users booking ≥ 2 seats and ≥ 1 room.
- *Note: has_children is included in an OR condition (TRUE or FALSE) to ensure the CASE doesn't break; it effectively ignores the child flag for FAMILY because the query author wanted to include married couples without kids.*
- "FLIGHT" vs "HOTEL": Defined strictly by exclusive booking behavior ($\text{total_flight_booked} > 0$ AND $\text{total_hotel_booked} = 0$, and vice versa). This explains why their discount usage percentages are 0 in the final output.

Known Limitations:

- Seasonality is not accounted for (data starts Jan 2024 only).
- Averages are used for multi-trip users; a user booking 1 business trip (last-minute) and 1 personal trip (3 months out) will be averaged, potentially misclassifying them.

Traveltide Segmentation Query

WITH

user_filter AS (

SELECT

u.user_id,

u.gender,

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    u.married,
    u.has_children,
    EXTRACT(YEAR FROM AGE(CURRENT_DATE, birthdate)) AS age
FROM users u
INNER JOIN sessions s ON u.user_id = s.user_id
WHERE s.session_start >= '2023-01-04'
GROUP BY u.user_id
HAVING COUNT(session_id) > 7 -- Focus on engaged, "reward-worthy" users
),
user_segments_parameter AS (
    SELECT
        usp.user_id,
        usp.gender,
        usp.married,
        usp.age,
        usp.has_children,
        COALESCE(SUM(changed_bags), 0) AS total_checked_bags,

-- Count successful trips (non-cancelled)
        COUNT(DISTINCT s.trip_id) FILTER (WHERE s.trip_id IS NOT NULL AND NOT s.cancellation) AS
total_trips,

-- Count cancelled trips
        COUNT(DISTINCT s.trip_id) FILTER (WHERE s.cancellation) AS cancelled_trips,

-- Discount usage counts
        COUNT(DISTINCT s.trip_id) FILTER (WHERE s.flight_discount AND NOT s.cancellation) AS
total_flight_discount,
        COUNT(DISTINCT s.trip_id) FILTER (WHERE s.hotel_discount AND NOT s.cancellation) AS
total_hotel_discount,

-- Return flight tracking

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COUNT(DISTINCT f.trip_id) FILTER (WHERE f.return_flight_booked AND NOT s.cancellation) AS
total_return_flight_booked,

-- Average discount amounts (only for trips where a discount was actually used)

COALESCE(ROUND(AVG(s.flight_discount_amount) FILTER (WHERE s.flight_discount AND
s.flight_booked AND s.trip_id IS NOT NULL AND NOT s.cancellation), 3), 0) AS
avg_flight_discount_amount,

COALESCE(ROUND(AVG(s.hotel_discount_amount) FILTER (WHERE s.hotel_discount AND
s.hotel_booked AND s.trip_id IS NOT NULL AND NOT s.cancellation), 3), 0) AS
avg_hotel_discount_amount,

-- Vertical booking counts

COUNT(DISTINCT s.trip_id) FILTER (WHERE s.hotel_booked AND NOT s.cancellation) AS
total_hotel_booked,

COUNT(DISTINCT s.trip_id) FILTER (WHERE s.flight_booked AND NOT s.cancellation) AS
total_flight_booked,

-- Behavioral averages (group size, length of stay)

COALESCE(ROUND(AVG(h.rooms) FILTER (WHERE s.hotel_booked AND s.trip_id IS NOT NULL AND
NOT s.cancellation), 0), 0) AS avg_rooms_booked,

COALESCE(ROUND(AVG(f.seats) FILTER (WHERE s.flight_booked AND s.trip_id IS NOT NULL AND
NOT s.cancellation), 0), 0) AS avg_seats_booked,

COALESCE(ROUND(AVG(h.nights) FILTER (WHERE s.hotel_booked AND s.trip_id IS NOT NULL AND
NOT s.cancellation), 0), 0) AS avg_nights_stay,

-- Days between session start and departure (proxy for "planning ahead" vs "last minute")

COALESCE(ROUND(AVG(DATE(f.departure_time) - DATE(s.session_start)) FILTER (WHERE s.trip_id
IS NOT NULL AND NOT s.cancellation), 0), 0) AS days_to_departure,

-- Day of Week (0=Sun, 6=Sat). Used to identify weekenders.

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        COALESCE(ROUND(AVG(EXTRACT(DOW FROM f.departure_time)) FILTER (WHERE s.trip_id IS NOT
NULL AND NOT s.cancellation), 0), 0) AS avg_day_of_departure

FROM user_filter usp
LEFT JOIN sessions s ON usp.user_id = s.user_id
LEFT JOIN hotels h ON h.trip_id = s.trip_id
LEFT JOIN flights f ON f.trip_id = s.trip_id
WHERE s.session_start >= '2023-01-04'
GROUP BY usp.user_id, usp.gender, usp.married, usp.age, usp.has_children
),
segment_detail AS (
SELECT
    usp.user_id,
    total_trips,
    cancelled_trips,
    avg_nights_stay,
    avg_seats_booked,
    avg_rooms_booked,
    total_hotel_booked,
    total_flight_booked,
    total_hotel_discount,
    total_flight_discount,
    total_return_flight_booked,
    avg_hotel_discount_amount,
    avg_flight_discount_amount,
    total_checked_bags,
    days_to_departure,
    avg_day_of_departure,
    CASE WHEN usp.total_checked_bags > 0 THEN TRUE ELSE FALSE END AS extra_checked_bags,

-- SEGMENT CLASSIFICATION LOGIC (Priority Order Matters!)
CASE

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-- 1. NEW: No successful trips yet. They are "window shoppers".

WHEN usp.total_trips = 0 OR usp.total_trips = usp.cancelled_trips THEN 'NEW'

-- 2. WEEKENDER: Departing Fri/Sat (DOW 5,6) and staying short (1-3 nights).

WHEN usp.avg_day_of_departure IN (5, 6) AND usp.avg_nights_stay BETWEEN 1 AND 3 THEN
'WEEKENDER'

-- 3. FAMILY: Married, booking >=2 seats and >=1 room.

-- (has_children TRUE/FALSE included to avoid breaking the CASE, but effectively ignored).

WHEN usp.married = TRUE AND usp.avg_rooms_booked >= 1 AND usp.avg_seats_booked >= 2
THEN 'FAMILY'

-- 4. GROUP: 2+ seats/rooms or 1 seat with 2+ rooms (friends/family).

WHEN (usp.avg_rooms_booked >= 1 AND usp.avg_seats_booked >= 2) OR
(usp.avg_seats_booked = 1 AND usp.avg_rooms_booked >= 2) THEN 'GROUP'

-- 5. BUSINESS: Solo traveler, weekday OR weekend long-stay, booked <15 days out (urgency).

WHEN (usp.avg_rooms_booked = 1 AND usp.avg_seats_booked = 1)
AND ((usp.avg_day_of_departure IN (0,1,2,3,4) AND usp.avg_nights_stay <= 3)
OR (usp.avg_day_of_departure IN (5,6) AND usp.avg_nights_stay > 3))
AND usp.days_to_departure < 15 THEN 'BUSINESS'

-- 6. SOLO: Catch-all for solo travelers not fitting the business logic.

WHEN (usp.avg_rooms_booked = 1 AND usp.avg_seats_booked = 1) THEN 'SOLO'

-- 7. FLIGHT: Exclusive flight bookers (zero hotels).

WHEN (usp.total_flight_booked > 0 AND usp.total_hotel_booked = 0) THEN 'FLIGHT'

-- 8. HOTEL: Exclusive hotel bookers (zero flights).

WHEN usp.total_hotel_booked > 0 AND usp.total_flight_booked = 0 THEN 'HOTEL'

ELSE 'NA'

END AS segment

FROM user_segments_parameter usp

)

-- Final Aggregation

SELECT DISTINCT

sd.segment,

COUNT(sd.user_id) AS total_users,

ROUND(100.0 * COUNT(DISTINCT sd.user_id) / NULLIF(SUM(COUNT(DISTINCT sd.user_id)) OVER (), 0), 2) AS user_pct,

COALESCE(ROUND(100.0 * SUM(sd.total_flight_discount) / NULLIF(SUM(sd.total_flight_booked), 0), 2), 0) AS flight_discount_usage_pct,

COALESCE(ROUND(AVG(sd.avg_flight_discount_amount) FILTER (WHERE sd.avg_flight_discount_amount > 0), 3), 0) AS avg_flight_discount,

COALESCE(ROUND(100.0 * SUM(sd.total_hotel_discount) / NULLIF(SUM(sd.total_hotel_booked), 0), 2), 0) AS hotel_discount_usage_pct,

COALESCE(ROUND(AVG(sd.avg_hotel_discount_amount) FILTER (WHERE sd.avg_hotel_discount_amount > 0), 3), 0) AS avg_hotel_discount,

COALESCE(ROUND(100.0 * SUM(sd.cancelled_trips) / NULLIF(SUM(sd.total_trips) + SUM(sd.cancelled_trips), 0), 2), 0) AS trip_cancel_pct,

ROUND(100.0 * COUNT(DISTINCT sd.user_id) FILTER (WHERE sd.extra_checked_bags) / NULLIF(COUNT(DISTINCT sd.user_id), 0), 2) AS pct_users_with_bags

FROM segment_detail sd

GROUP BY 1

ORDER BY total_users DESC; -- Added ORDER BY for readability

Query Result

	segment ^	total_users v	user_pct ^	flight_discount_usage_pct ^	avg_flight_discount ^	hotel_discount_usage_pct ^	avg_hotel_discount ^	trip_cancel_pct ^	pct_users_with_bags ^
1	BUSINESS	2229	37.16	15.37	0.142	13.99	0.112	2.10	80.89
2	SOLO	1299	21.66	18.32	0.137	15.59	0.111	3.45	78.52
3	GROUP	645	10.75	19.02	0.136	16.94	0.107	4.70	77.83
4	NEW	555	9.25	14.13	0.123	26.79	0.114	50.51	9.55
5	FAMILY	419	6.99	19.96	0.148	16.89	0.113	4.95	87.11
6	WEEKENDER	396	6.60	16.19	0.140	16.65	0.112	3.11	73.23
7	HOTEL	357	5.95	0	0	20.62	0.109	1.31	0.00
8	FLIGHT	98	1.63	17.09	0.148	0	0	7.87	63.27